NORTH BENGAL INTERNATIONAL UNIVERSITY

Lab Report : 3

TOPIC : WRITE A PROGRAM TO SHOW VARIABLES AS AN OUTPUT IN ASSEMBLY LANGUAGE

COURSE CODE : CSE 0611-3322

COURSE TITLE : MICROPROCESSOR & ASSEMBLY LANGUAGE SESSIONAL.

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Theory: Assembly language is a low-level programming language that's more readable than machine language. It uses symbolic names instead of binary code, making it easier to understand and write. Each instruction corresponds to a specific machine language instruction and is specific to a particular computer architecture. It's often used for tasks needing high performance or direct hardware control.

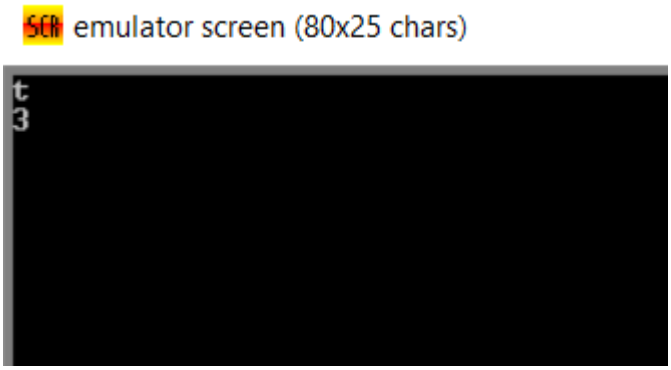
Show Variable Output in Assembly Language:

This assembly language program defines two variables, `var1` and `var2`. It sets up the data segment and displays the value of `var1`, followed by a newline. It then converts `var2` to its ASCII equivalent and displays it.

Code:

```
.model small
.stack 100h

.data
    var1 db 't'  ; Define a variable with the value 't'
    var2 db 3    ; Define a variable with the value 3

.code
main proc
    ; Initialize the data segment
    mov ax, @data
    mov ds, ax

    ; Display var1
    mov ah, 2
    mov dl, var1
    int 21h

    ; Print a newline (Line Feed and Carriage Return)
    mov ah, 2
    mov dl, 10
    int 21h
    mov dl, 13
    int 21h

    ; Convert var2 to ASCII and display it
    mov ah, 2
    add var2, 48  ; Convert integer to ASCII character
    mov dl, var2
    int 21h

exit:
    ; Exit program
    mov ah, 4ch
    int 21h
main endp
end main
```

Output:



Conclusion: This assembly language program defines two variables and displays their values using DOS interrupts. It includes handling for newline and carriage return characters to format the output. The program exits cleanly after displaying the values.